

A Feature-Driven Tensor Semantics for Minimalist Grammars

John Paulson

Dept. of Linguistics
University of Utah
john.paulson@utah.edu

Aniello De Santo

Dept. of Linguistics
University of Utah
aniello.desanto@utah.edu

Jon Rawski

Dept. of Linguistics and Language
Development
San Jose State University
jon.rawski@sjsu.edu



This version of the slides does not have animations; it was split automatically with PPspliT (<https://www.maxonthenet.altervista.org/ppsplit.php>). The version of the slides with animation (in pptx and ppt formats) can be found at https://github.com/JPJ280/jpj280.github.io/tree/master/publications/SCIL_2026_MG_Tensors/slides (see also the QR code at the end of the presentation).

Talk in one slide

This work bridges distributional approaches to semantics and “mainstream” generative grammar.

- We incorporate **Minimalist Grammars** into **vector-based semantic** models
- The resulting vector semantic models can **build the meaning of larger expressions compositionally**, by combining the meaning of smaller expressions
- This compositionality is **guided by the Minimalist Grammar syntax**

Distributionality and Compositionality

Distributionality and Compositionality

- Computational linguistics often makes use of *distributional* semantics, describing the meanings of expressions through co-occurrence in corpora

Distributionality and Compositionality

- Computational linguistics often makes use of *distributional* semantics, describing the meanings of expressions through *co-occurrence in corpora*
- Embedding the meanings of expressions into *vector spaces* is a popular way of implementing distributional models

Distributionality and Compositionality

- Computational linguistics often makes use of *distributional* semantics, describing the meanings of expressions through *co-occurrence in corpora*
- Embedding the meanings of expressions into *vector spaces* is a popular way of implementing distributional models
- In contrast, traditional/formal semantics research often analyzes meaning *compositionally*, *building up the meaning* of expressions from subparts

Distributionality and Compositionality

- Computational linguistics often makes use of *distributional* semantics, describing the meanings of expressions through *co-occurrence in corpora*
- Embedding the meanings of expressions into *vector spaces* is a popular way of implementing distributional models
- In contrast, traditional/formal semantics research often analyzes meaning *compositionally*, *building up the meaning* of expressions from subparts
- An approach that uses *vector space operations to combine the meanings* of smaller expressions into meanings for larger expressions can bridge these two approaches

Formal Grammars and Distributional Semantics

- In particular, having composition be **guided by a formal grammar** can bridge semantic and syntactic approaches, motivating how such operations should be applied

Formal Grammars and Distributional Semantics

- In particular, having composition be **guided by a formal grammar** can bridge semantic and syntactic approaches, motivating how such operations should be applied
- There is **already successful work** integrating Combinatory Categorical Grammars (CCGs; Steedman, 2001; Steedman and Baldridge, 2011) with distributional semantics in this way (Coecke et al., 2010; Maillard et al., 2014)

Formal Grammars and Distributional Semantics

- In particular, having composition be **guided by a formal grammar** can bridge semantic and syntactic approaches, motivating how such operations should be applied
- There is **already successful work** integrating Combinatory Categorical Grammars (CCGs; Steedman, 2001; Steedman and Baldridge, 2011) with distributional semantics in this way (Coecke et al., 2010; Maillard et al., 2014)
- This work **integrates such a model distributional semantics with Minimalist Grammars** (MGs; Stabler, 1996), which formalize the (broad strokes) syntactic approach of the Minimalist Program

Why MGs?

Why MGs?

- MGs are a mildly context-sensitive grammar formalism which are thought to be able to capture all of natural language syntax

Why MGs?

- MGs are a mildly context-sensitive grammar formalism which are thought to be able to capture all of natural language syntax
- MGs formalize “Minimalist Program-like” syntax
 - In this way, incorporating distributional semantics into MGs provides a link to “mainstream” syntactic theory

Why MGs?

- MGs are a mildly context-sensitive grammar formalism which are thought to be able to capture all of natural language syntax
- MGs formalize “Minimalist Program-like” syntax
 - In this way, incorporating distributional semantics into MGs provides a link to “mainstream” syntactic theory
- MGs have also been useful in analyzing sentence processing (Kobele et al., 2012; De Santo, 2020, 2025)

Why MGs?

- MGs are a mildly context-sensitive grammar formalism which are thought to be able to capture all of natural language syntax
- MGs formalize “Minimalist Program-like” syntax
 - In this way, incorporating distributional semantics into MGs provides a link to “mainstream” syntactic theory
- MGs have also been useful in analyzing sentence processing (Kobele et al., 2012; De Santo, 2020, 2025)
- There is also a vector-based analysis for the syntax of MGs (beim Graben & Gerth, 2012)

Why MGs?

- MGs are a mildly context-sensitive grammar formalism which are thought to be able to capture all of natural language syntax
- MGs formalize “Minimalist Program-like” syntax
 - In this way, incorporating distributional semantics into MGs provides a link to “mainstream” syntactic theory
- MGs have also been useful in analyzing sentence processing (Kobele et al., 2012; De Santo, 2020, 2025)
- There is also a vector-based analysis for the syntax of MGs (beim Graben & Gerth, 2012)
- It's not just CCGs!

Minimalist Grammars

Minimalist Grammars

- Trees built from a lexicon and the operations Merge and Move
- Each tree (including lexical items) has a string of features dictating which operations can be used
 - Merge features:
 - Selectee features (f)
 - Selector features (=f)
 - Move features:
 - Licensee features (f)
 - Licensor features (f⁺)

Lexical Item	Feature String
<i>what</i>	[D, wh ⁺]
<i>eat</i>	[=D, V]
$\emptyset_v$ (little v)	[=V, =D, v]
<i>Alice</i>	[D, nom ⁻]
$\emptyset_T$ (T head)	[=v, nom ⁺ , T]
<i>did</i>	[=T, wh ⁺ , C]

Minimalist Grammars

Merge is a (partial) binary operation. It takes one tree with a selectee feature (f) and one with a matching selector feature ($=f$), combines them into one tree, checking (deleting) both features

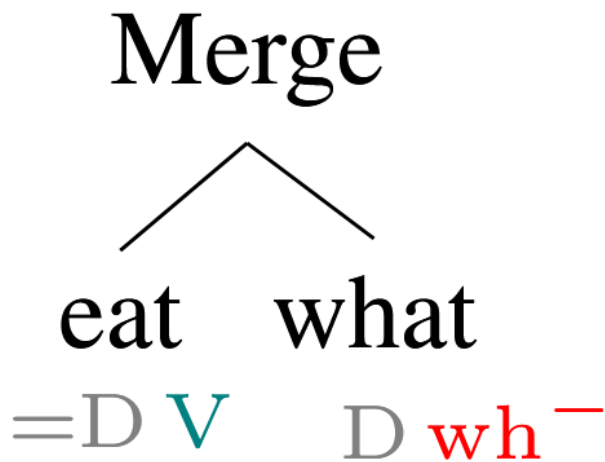
Minimalist Grammars

Merge is a (partial) binary operation. It takes one tree with a selectee feature (f) and one with a matching selector feature (=f), combines them into one tree, checking (deleting) both features

eat what
=D V D wh⁻

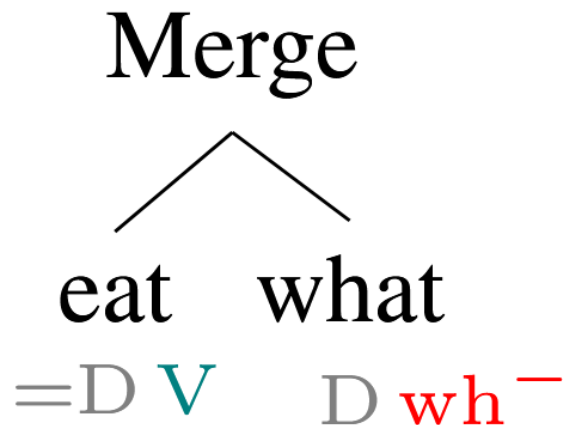
Minimalist Grammars

Merge is a (partial) binary operation. It takes one tree with a selectee feature (f) and one with a matching selector feature (=f), combines them into one tree, checking (deleting) both features

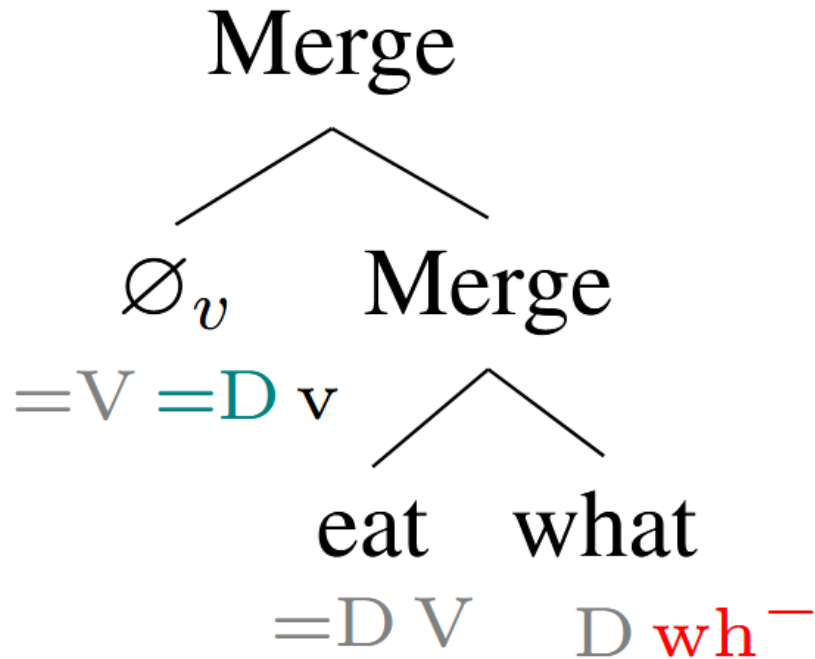


Minimalist Grammars

$$\emptyset_v$$
$$=V =D \text{ } v$$



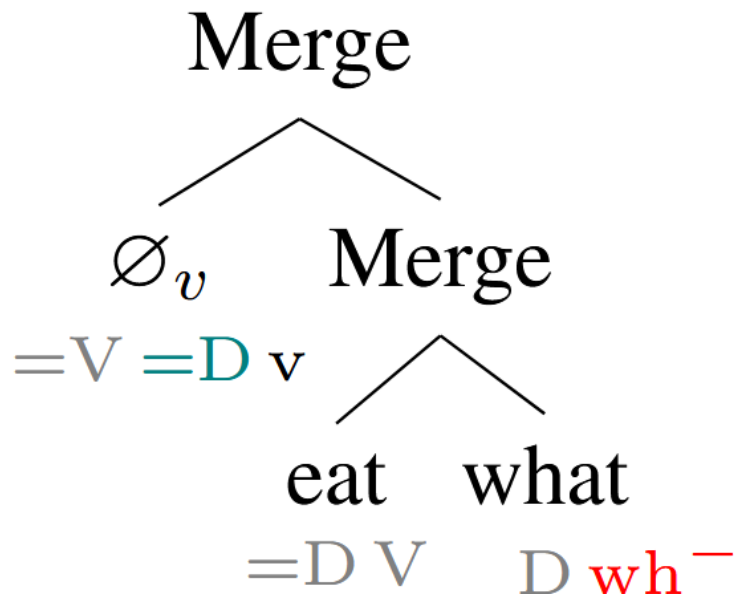
Minimalist Grammars



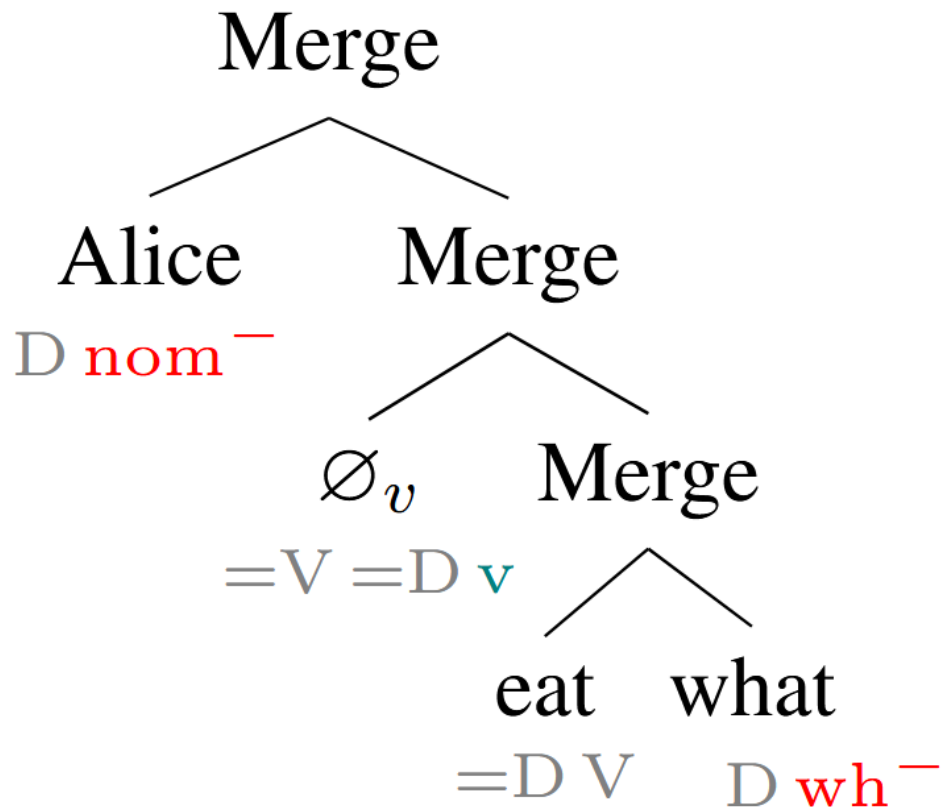
Minimalist Grammars

Alice

D_{nom}^-



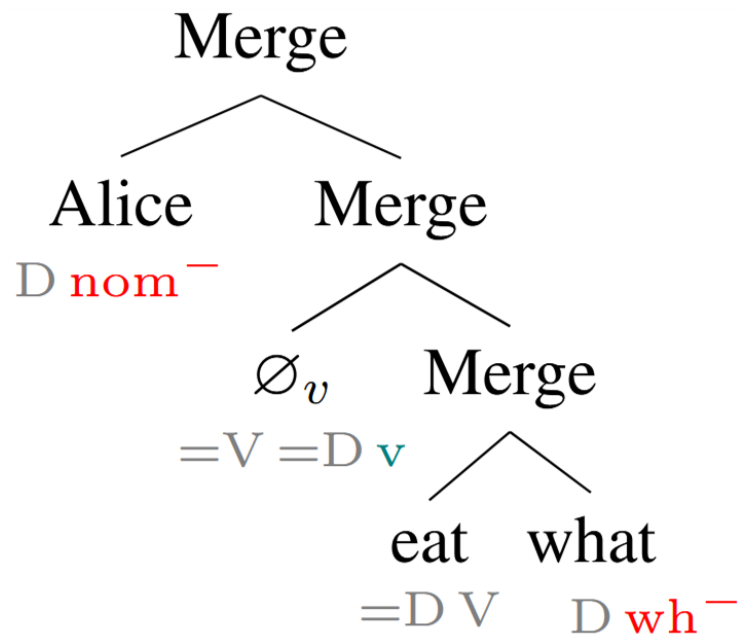
Minimalist Grammars



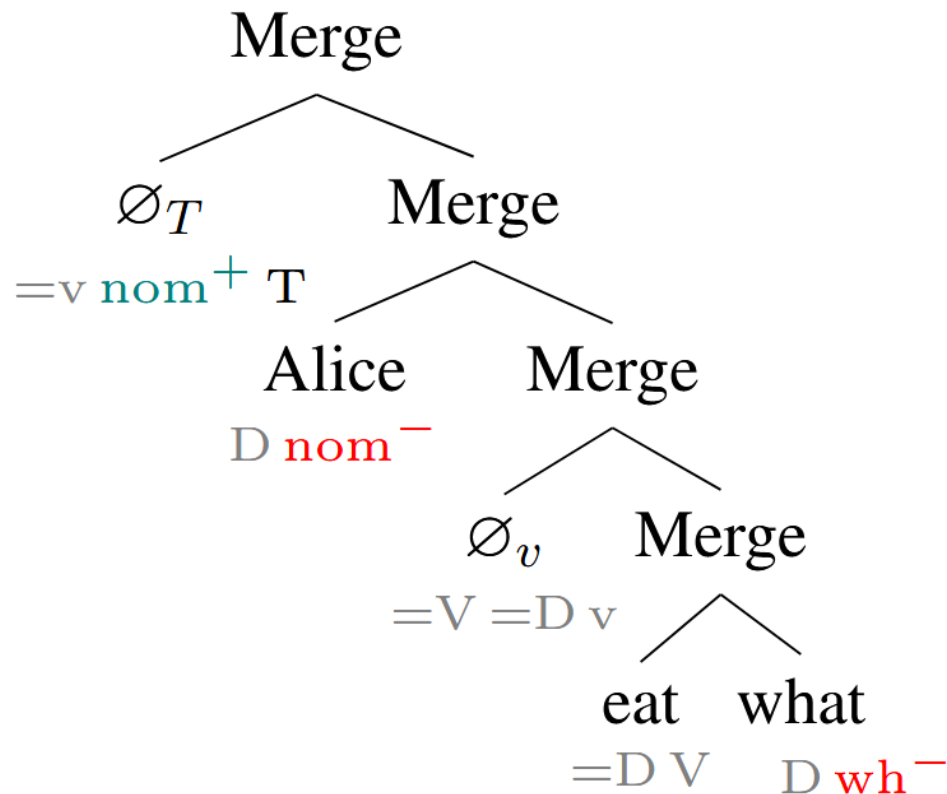
Minimalist Grammars

$$\emptyset_T$$

$$=v \text{ nom}^+ T$$



Minimalist Grammars

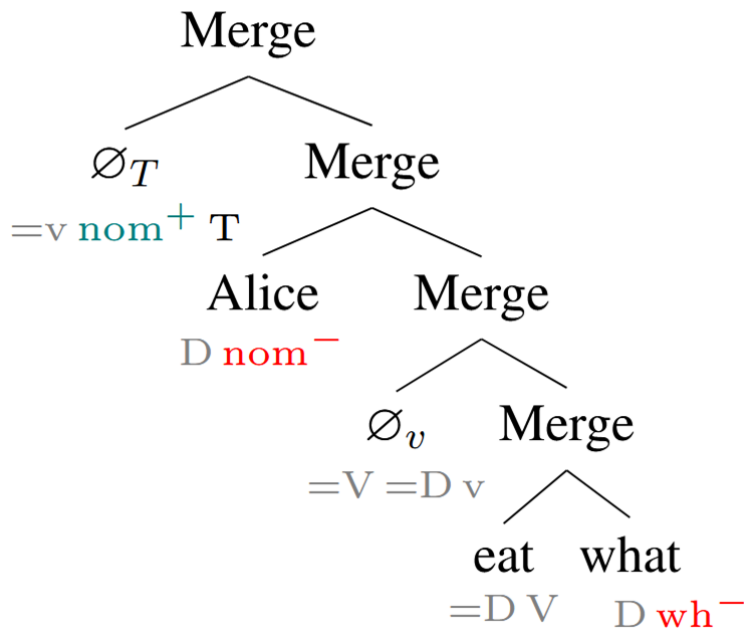


Minimalist Grammars

Move is a (partial) unary operation. It takes a tree with an active licensor (f^+) feature and *exactly one* matching licensee feature (f) and checks (deletes) both.

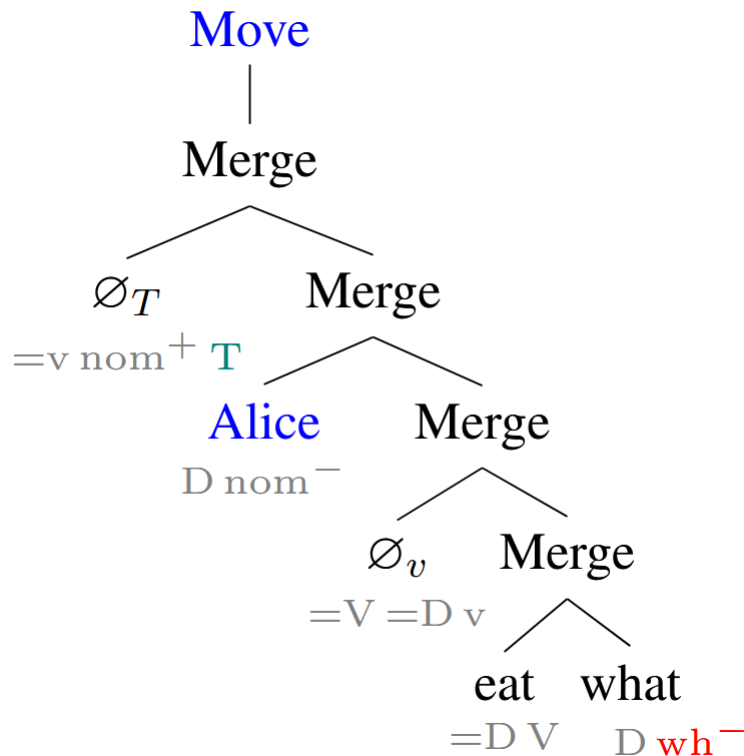
Minimalist Grammars

Move is a (partial) unary operation. It takes a tree with an active licenser (f^+) feature and *exactly one* matching licensee feature (f) and checks (deletes) both.



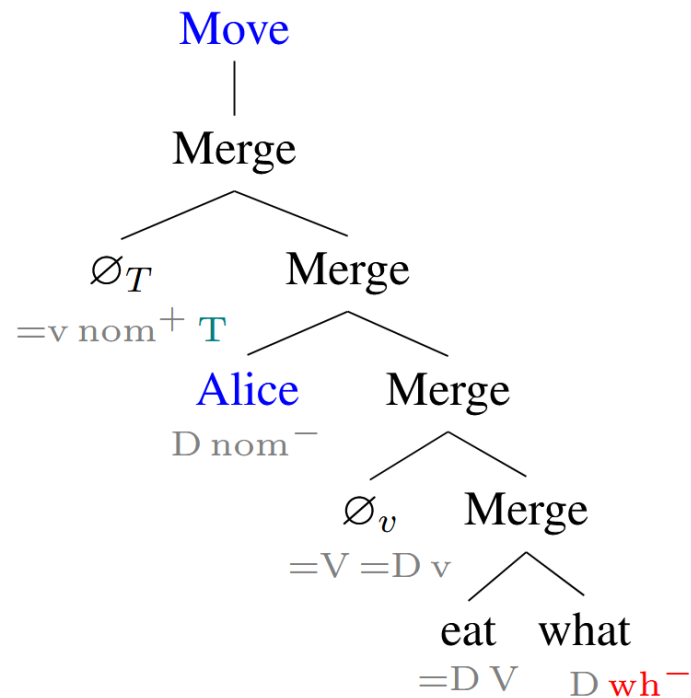
Minimalist Grammars

Move is a (partial) unary operation. It takes a tree with an active licenser (f^+) feature and *exactly one* matching licensee feature (f) and checks (deletes) both.

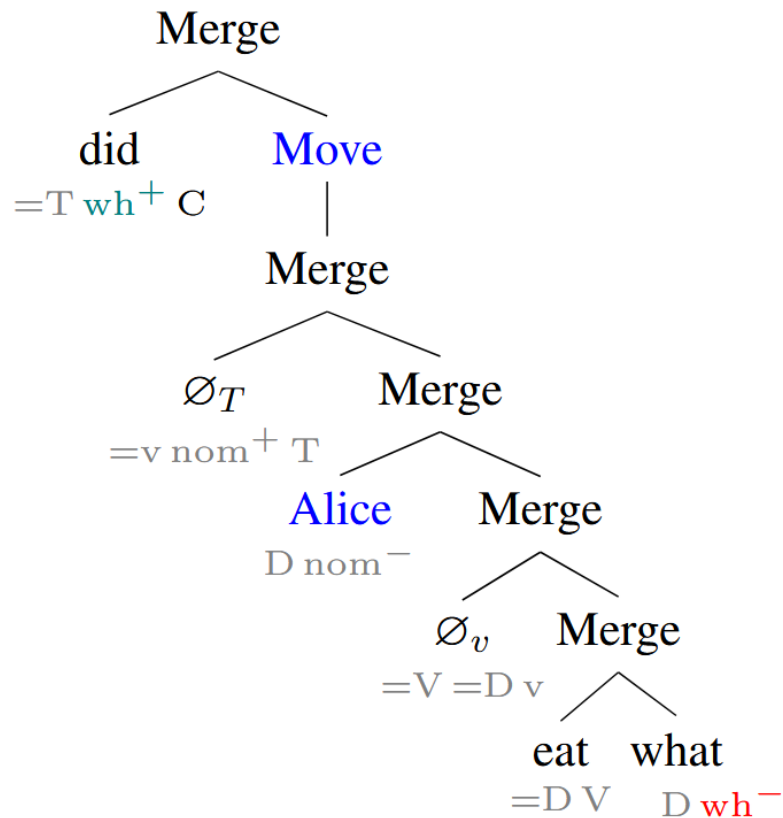


Minimalist Grammars

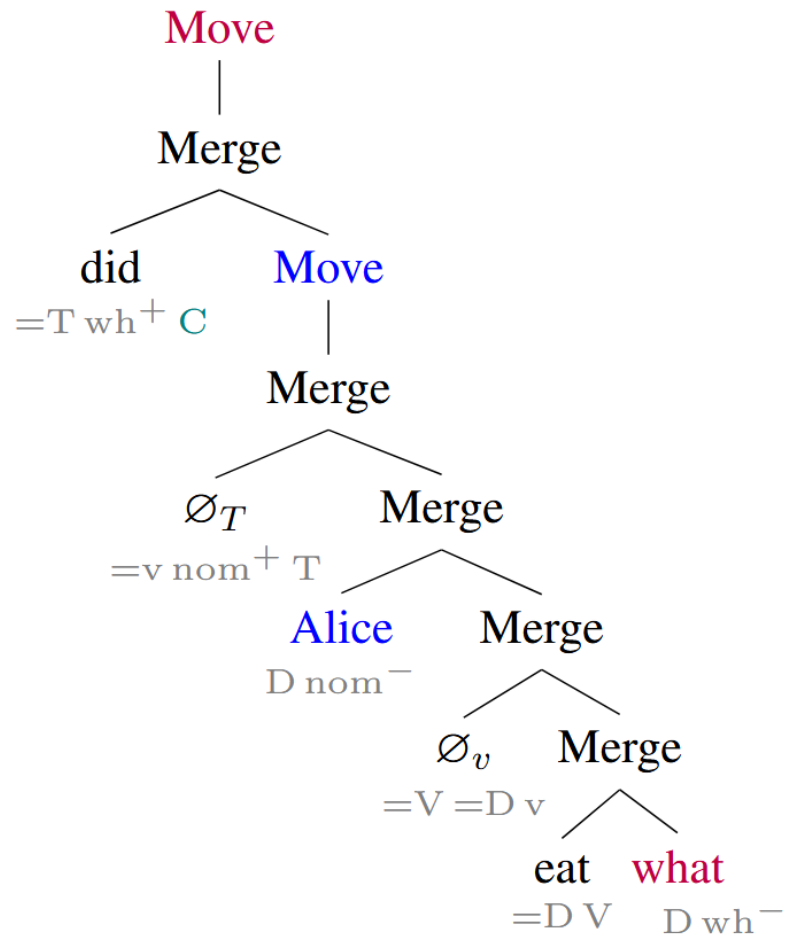
did
 $=T_{wh}^+ C$



Minimalist Grammars

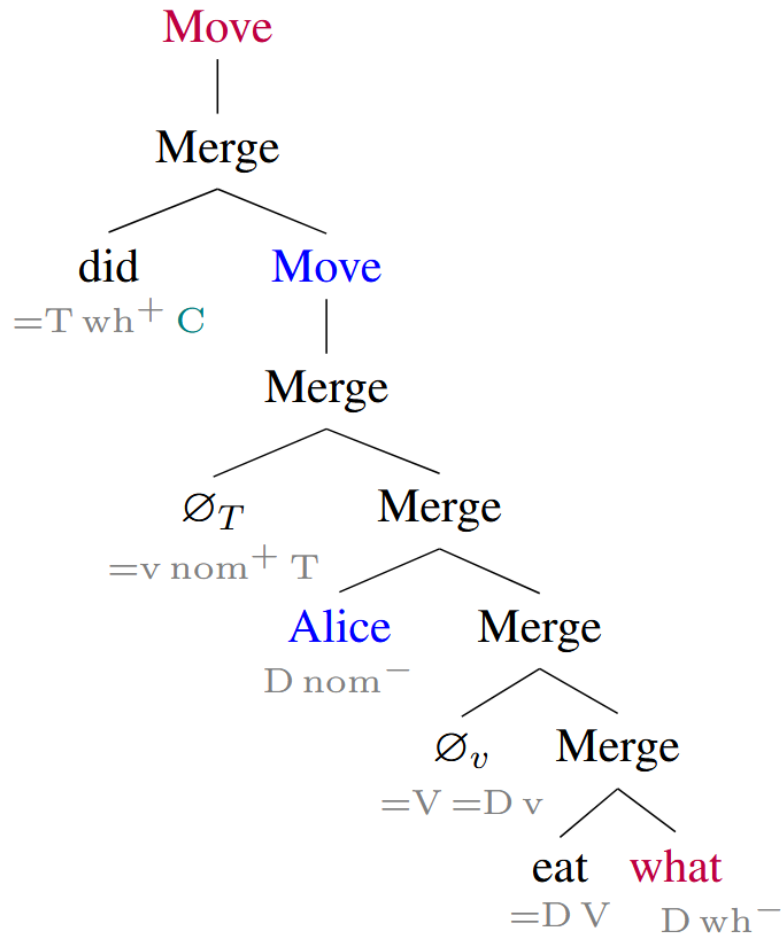


Minimalist Grammars



Minimalist Grammars

Only feature remaining is C, so this tree is accepted by the grammar.

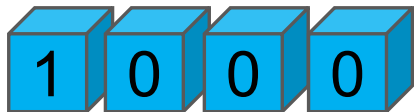


Tensor Semantics for MGs

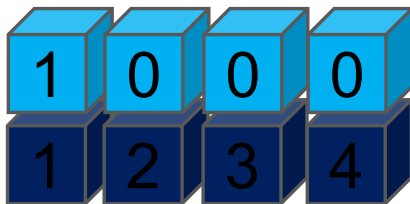
Tensors

- An order- k tensor is a k -dimensional grid of numbers
- An order- k tensor is a **vector in the tensor product** $V_1 \otimes \cdots \otimes V_k$ of k (finite dimensional) vector spaces $V_1, \dots, V_k$

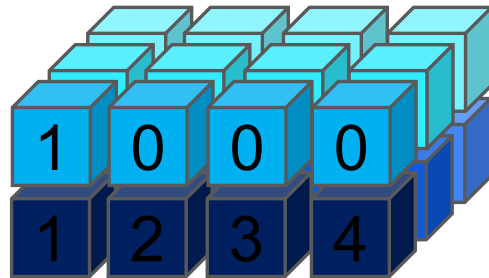
Order 1 tensor (vector)



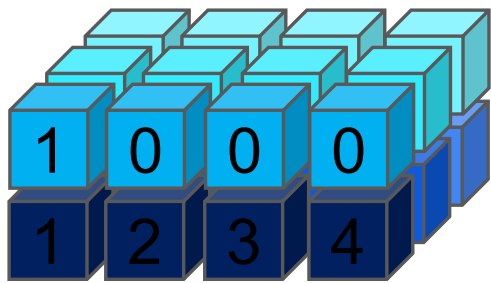
Order 2 tensor (matrix)



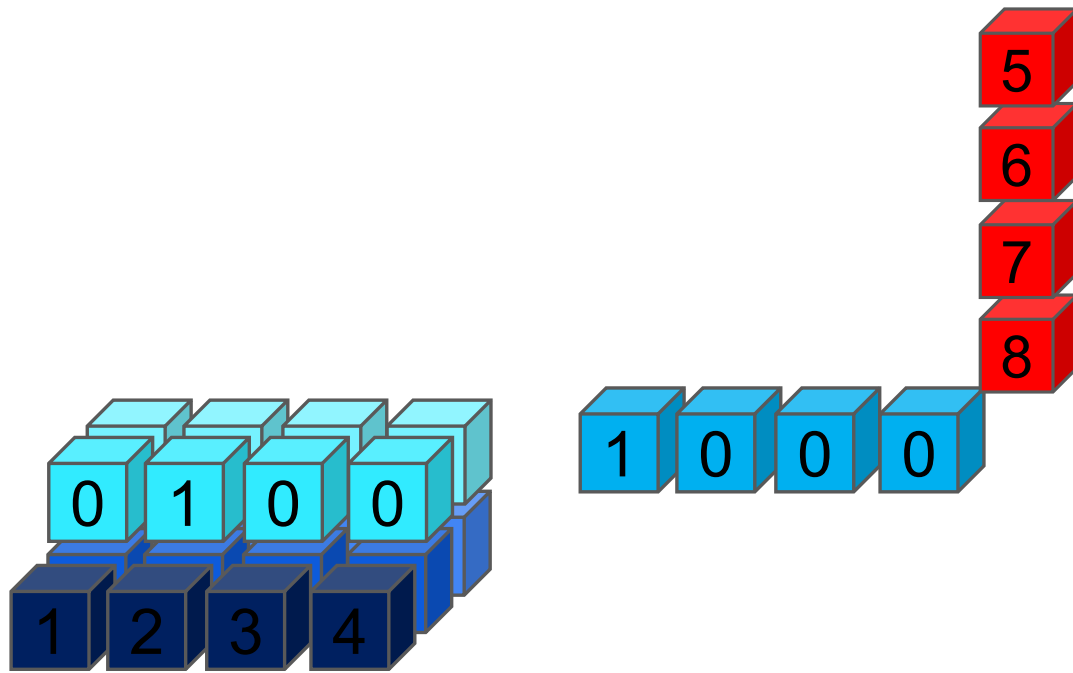
Order 3 tensor



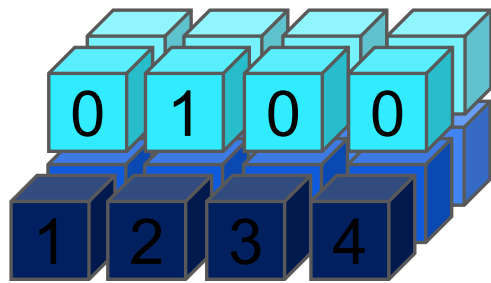
Tensor Contraction



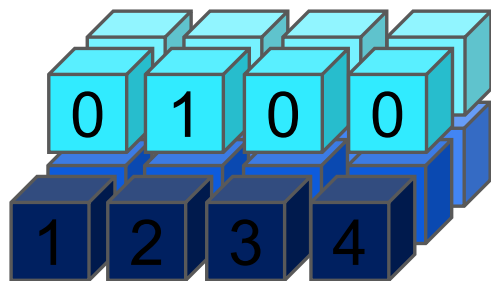
Tensor Contraction



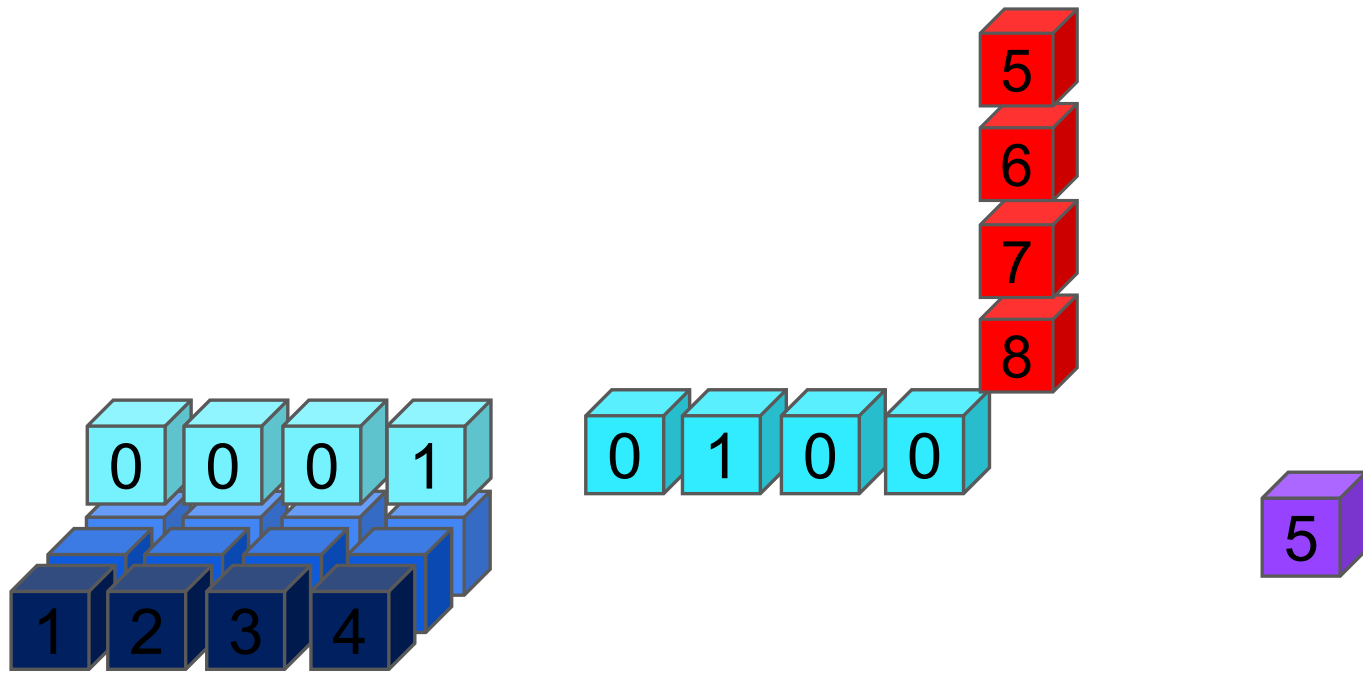
Tensor Contraction



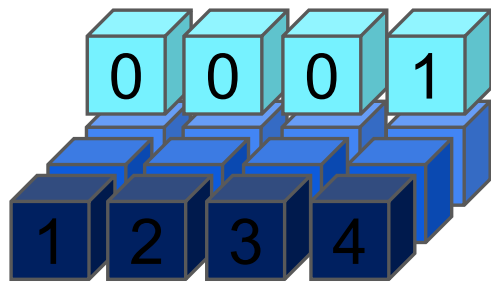
Tensor Contraction



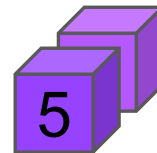
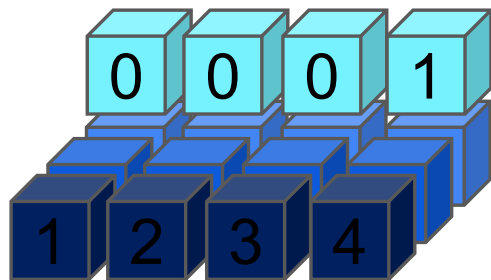
Tensor Contraction



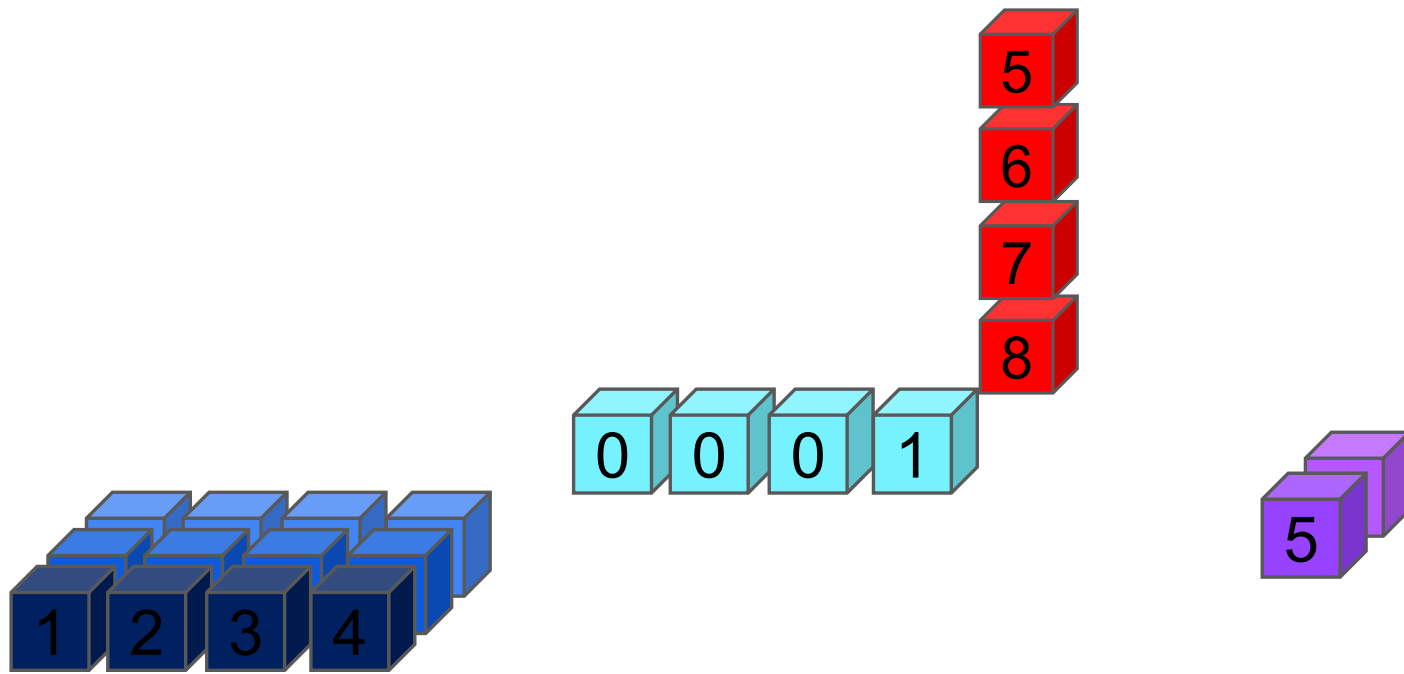
Tensor Contraction



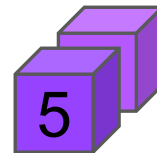
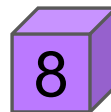
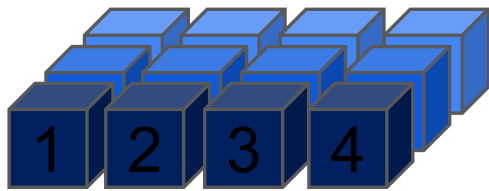
Tensor Contraction



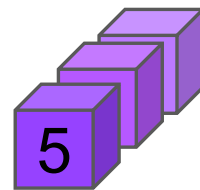
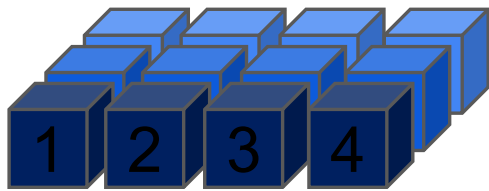
Tensor Contraction



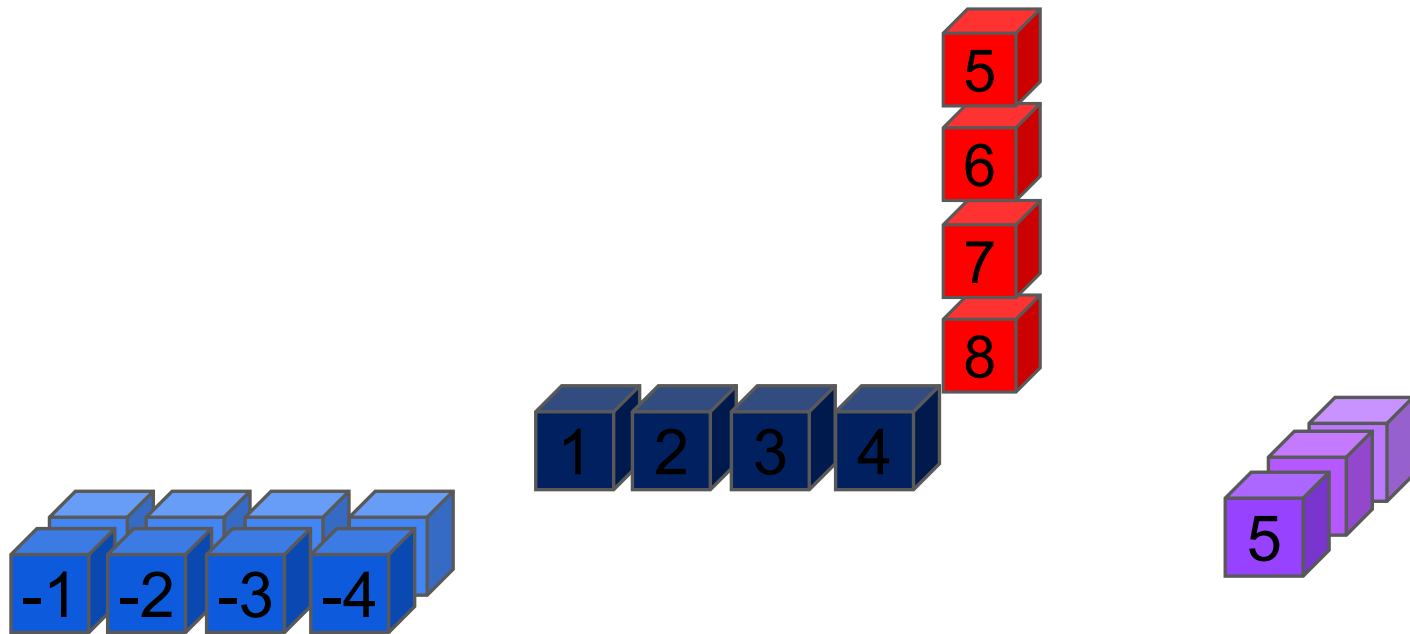
Tensor Contraction



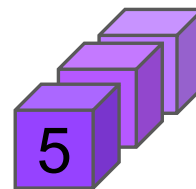
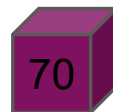
Tensor Contraction



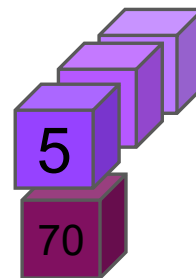
Tensor Contraction



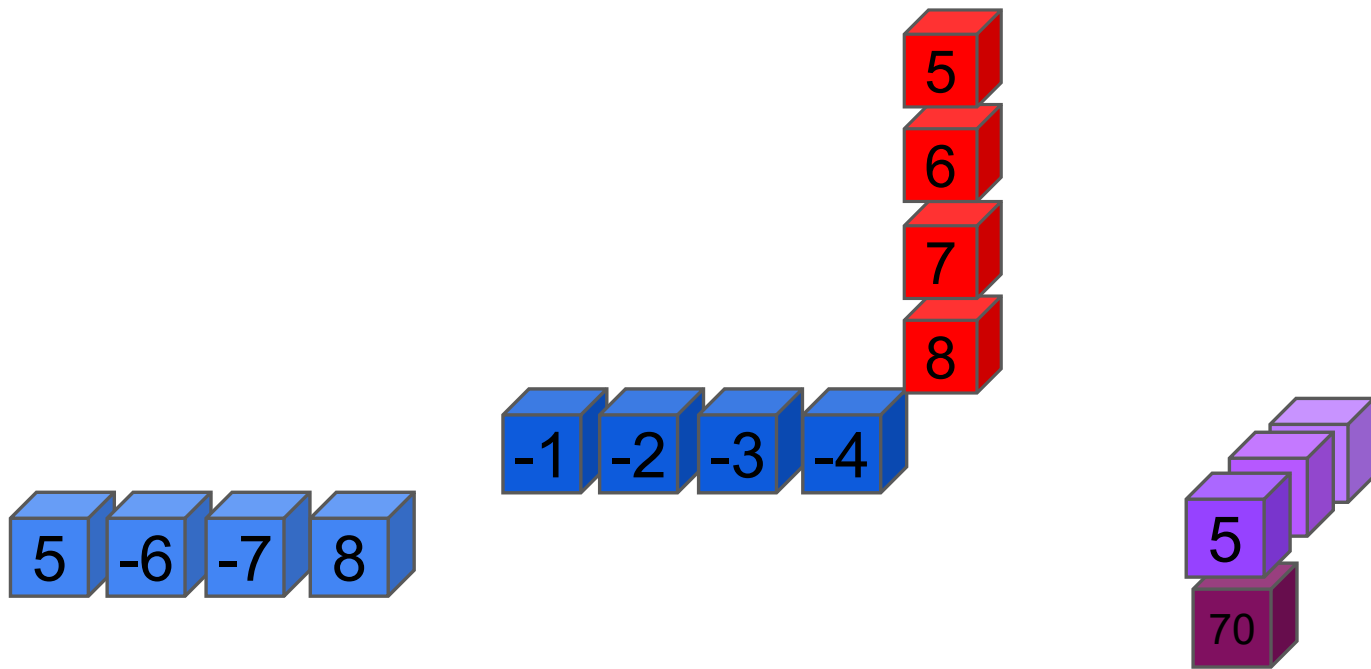
Tensor Contraction



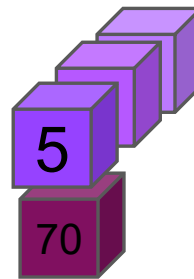
Tensor Contraction



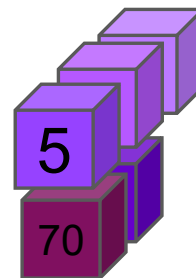
Tensor Contraction



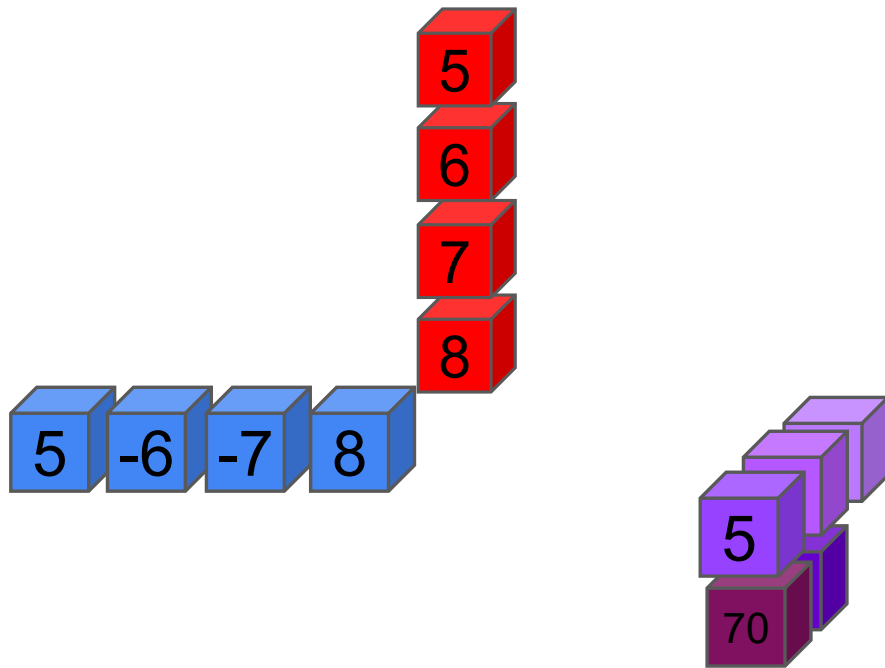
Tensor Contraction



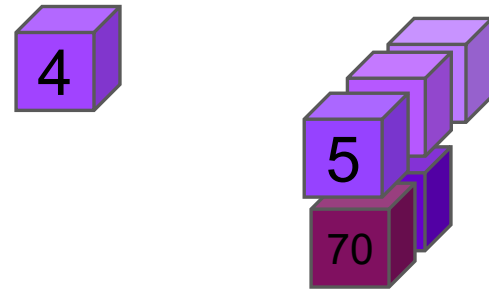
Tensor Contraction



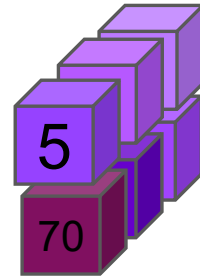
Tensor Contraction



Tensor Contraction



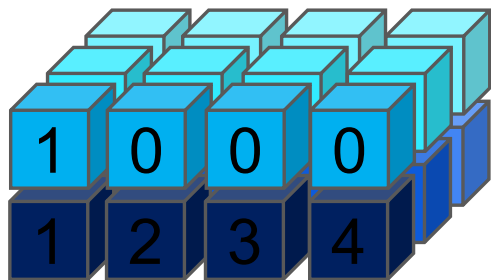
Tensor Contraction



Tensor Contraction

$$\mathbb{R}^2 \otimes \mathbb{R}^3 \otimes \mathbb{R}^4$$

Ψ



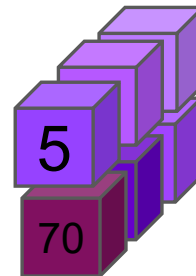
$$\mathbb{R}^4$$

Ψ



$$\mathbb{R}^2 \otimes \mathbb{R}^3$$

Ψ



Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

[D wh⁻]

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

D

[**D** wh⁻]

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

what

D

[**D** wh⁻]

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

$\Phi(\textit{what}) \in \mathbf{D}$



what

[**D** wh⁻]

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

$\Phi(\textit{what}) \in \mathbf{D}$



what

[**D** wh⁻]

[=D V]

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

$\Phi(\textit{what}) \in \mathbf{D}$



what

$[\mathbf{D} \text{ wh}^-]$

V

$[=D \mathbf{V}]$

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

$$\Phi(\textit{what}) \in \mathbf{D}$$



what

$$[\mathbf{D} \quad \textit{wh}^-]$$

$$\mathbf{V} \otimes \mathbf{D}$$

$$[=\mathbf{D} \quad \mathbf{V}]$$

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

$\Phi(\textit{what}) \in \mathbf{D}$

$\uparrow$
what

$[\mathbf{D} \quad \textit{wh}^-]$

$\mathbf{V} \otimes \mathbf{D}$

eat $[=\mathbf{D} \quad \mathbf{V}]$

Tensor Semantics for MGs

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

$$\begin{array}{c}
 \Phi(\textit{what}) \in \mathbf{D} \\
 \uparrow \\
 \textit{what} \quad [\mathbf{D} \quad \textit{wh}^-] \\
 \\
 \Phi(\textit{eat}) \in \mathbf{V} \otimes \mathbf{D} \\
 \uparrow \\
 \textit{eat} \quad [=\mathbf{D} \quad \mathbf{V}]
 \end{array}$$

Tensor Semantics for MGs

$\mathbf{V} \otimes \mathbf{D}$

Ψ

$\Phi(\textit{eat})$



eat

$\mathbf{D}$

Ψ

$\Phi(\textit{what})$



what

Tensor Semantics for MGs

$\mathbf{V} \otimes \mathbf{D}$

Ψ

$\Phi(eat)$



eat

$\mathbf{D}$

Ψ

$\Phi(what)$



what

merge(eat, what)

Tensor Semantics for MGs

$\mathbf{V} \otimes \mathbf{D}$

Ψ

$\Phi(eat)$



eat

$\mathbf{D}$

Ψ

$\Phi(what)$



what

$\mathbf{V}$

Ψ

merge(eat, what)

Tensor Semantics for MGs

V $\otimes$ **D**

Ψ

$\Phi(eat)$



eat

D

Ψ

$\Phi(what)$



what

V

Ψ

$\Phi(eat) \cdot \Phi(what)$



merge(eat, what)

Tensor Semantics for MGs

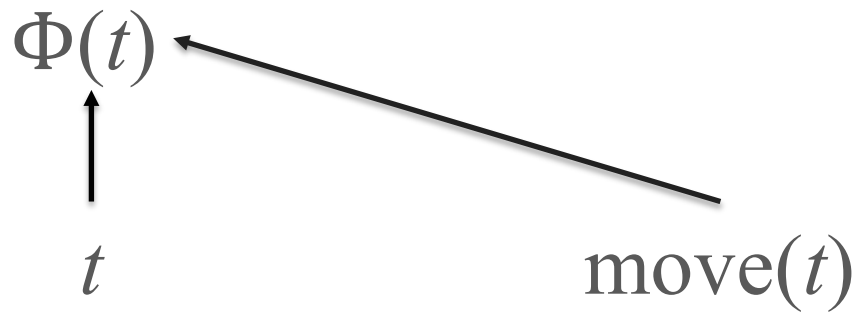
$$\begin{array}{c} \Phi(t) \\ \uparrow \\ t \end{array}$$

Tensor Semantics for MGs

$$\begin{array}{c} \Phi(t) \\ \uparrow \\ t \end{array}$$

$\text{move}(t)$

Tensor Semantics for MGs



Tensor Semantics for MGs

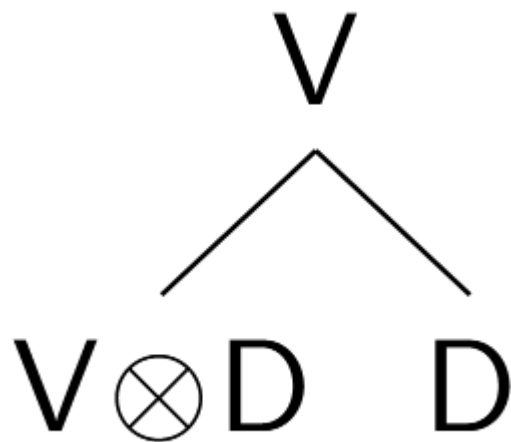
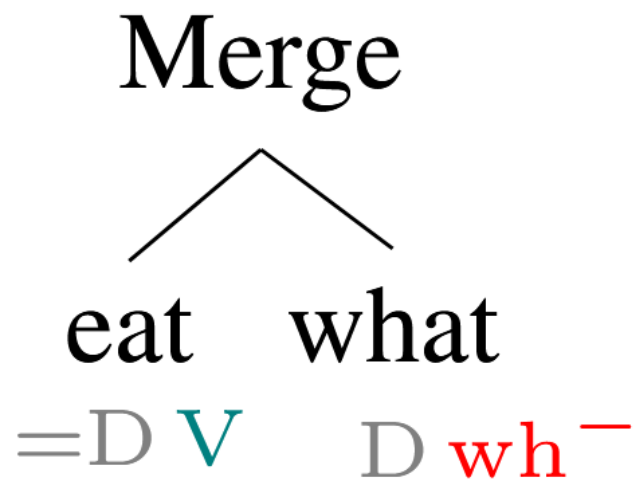
$$\Phi(\text{merge}(t_1, t_2)) = \Phi(t_1) \cdot \Phi(t_2)$$

$$\Phi(\text{move}(t)) = \Phi(t)$$

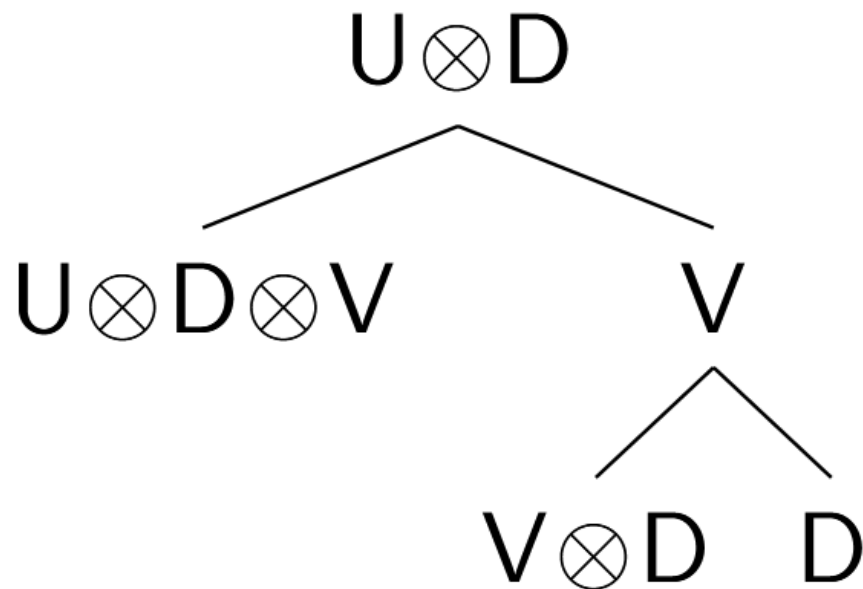
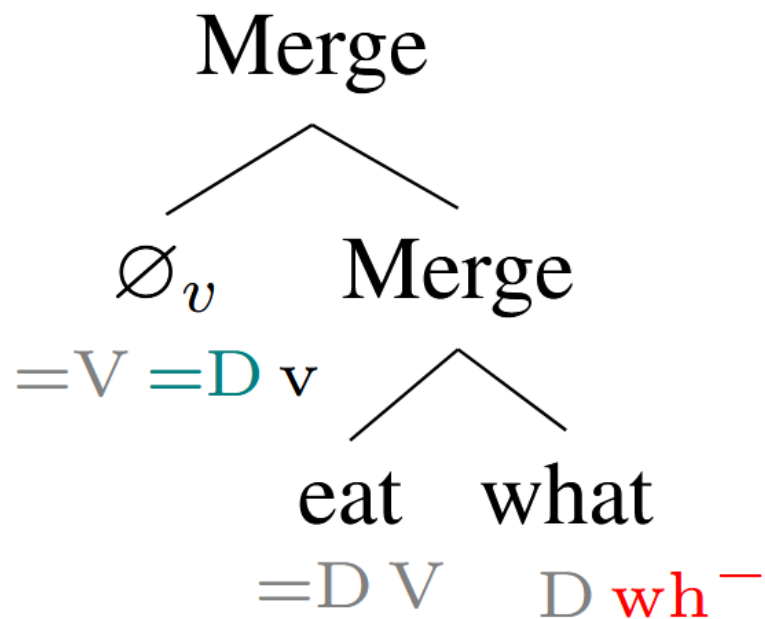
Example

Lexical Item	Feature String	Space of Tensor
<i>what</i>	$[D, wh^+]$	D
<i>eat</i>	$[=D, V]$	$V \otimes D$
$\emptyset_v$ (little v)	$[=V, =D, v]$	$U \otimes D \otimes V$
<i>Alice</i>	$[D, nom^-]$	D
$\emptyset_T$ (T head)	$[=v, nom^+, T]$	$T \otimes U$
<i>did</i>	$[=T, wh^+, C]$	$C \otimes T$

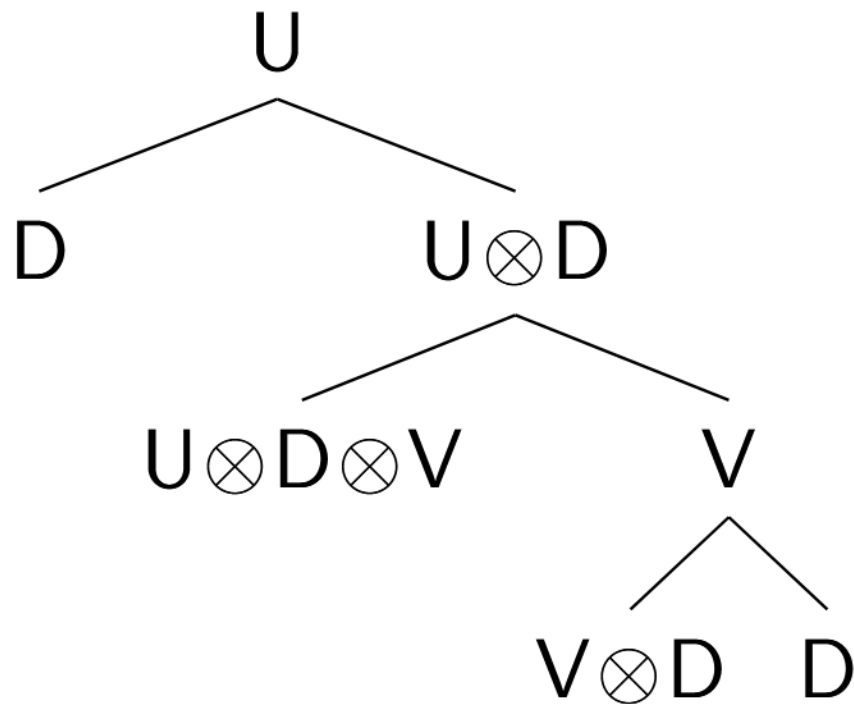
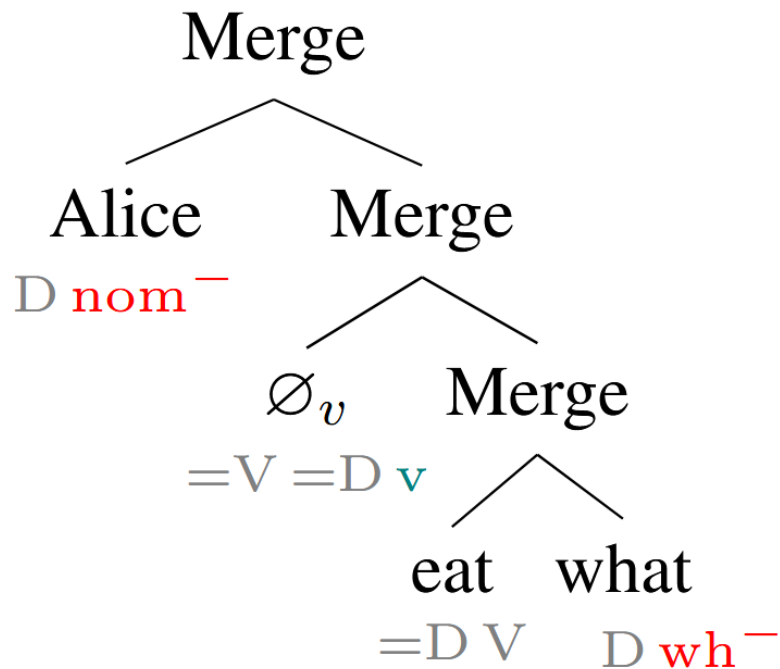
Example



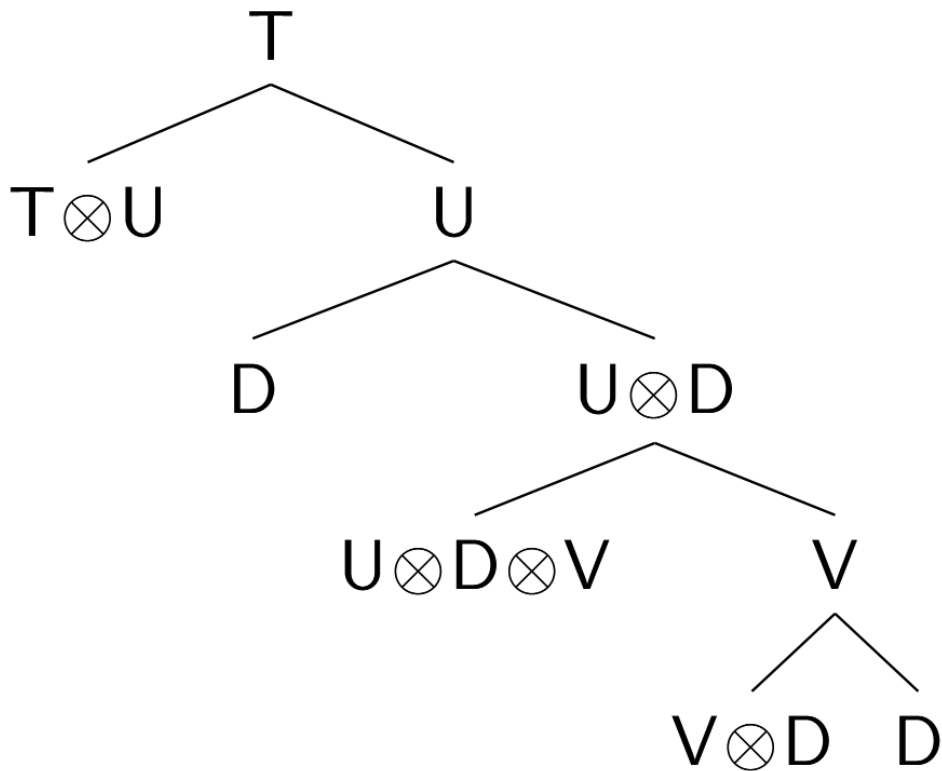
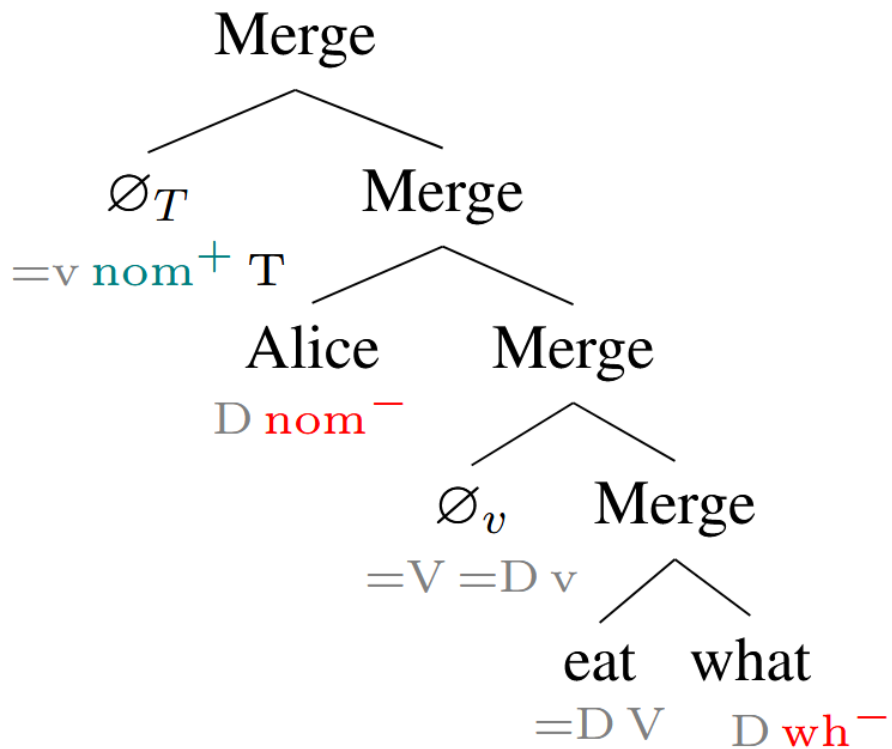
Example



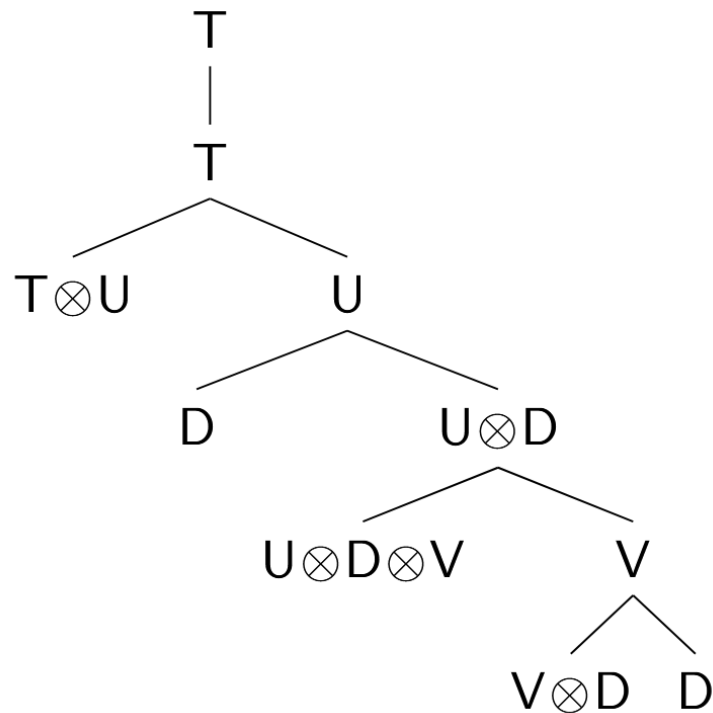
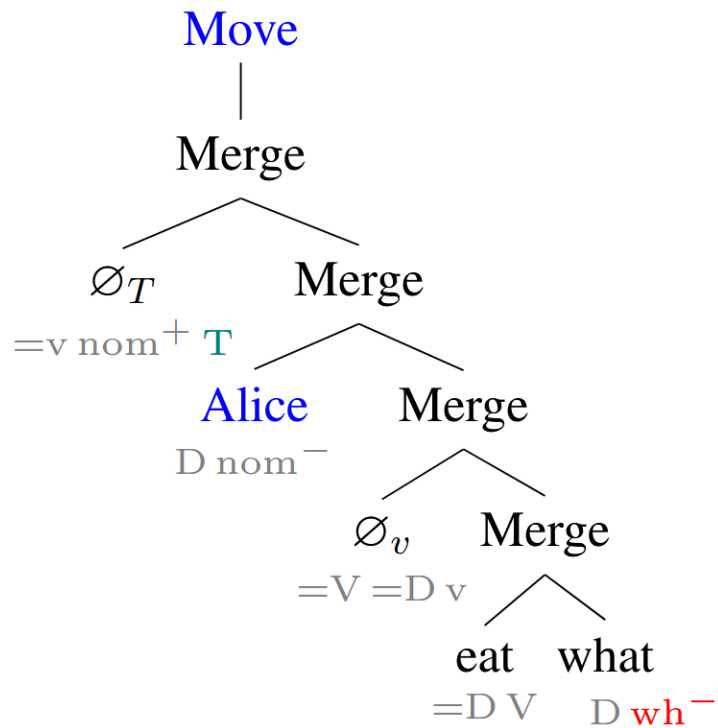
Example



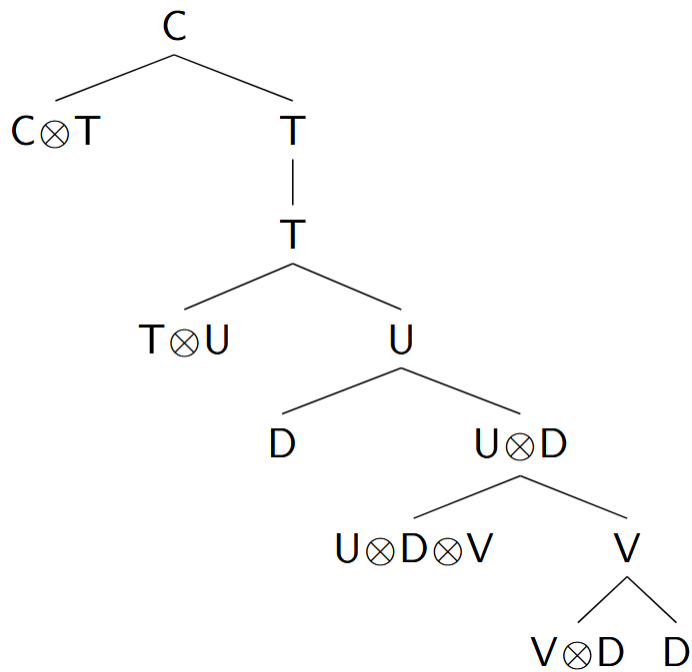
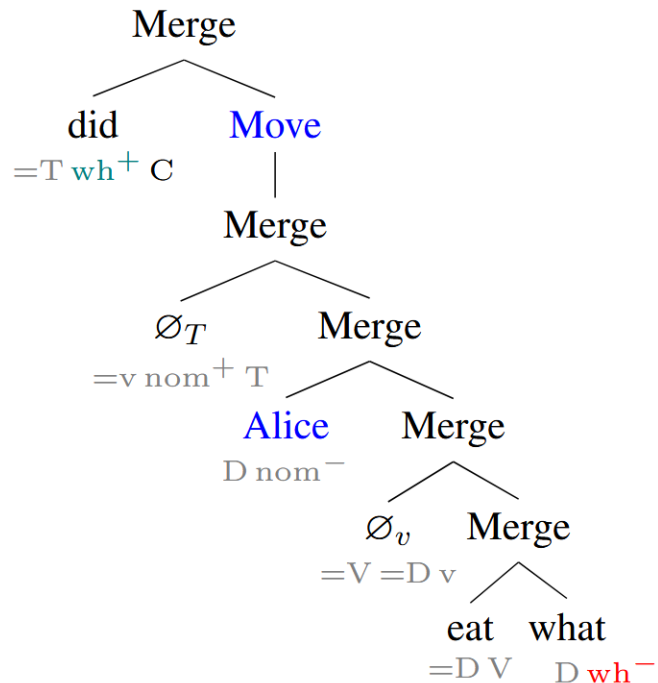
Example



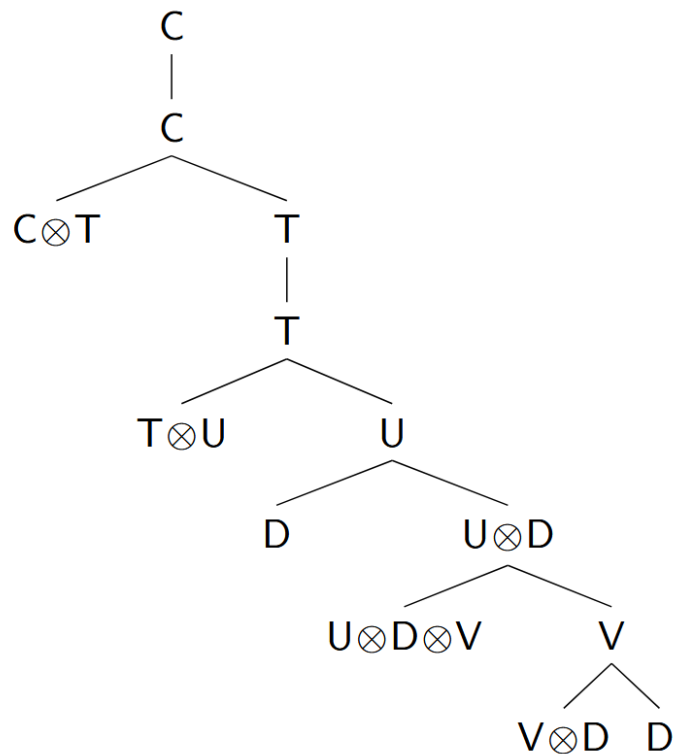
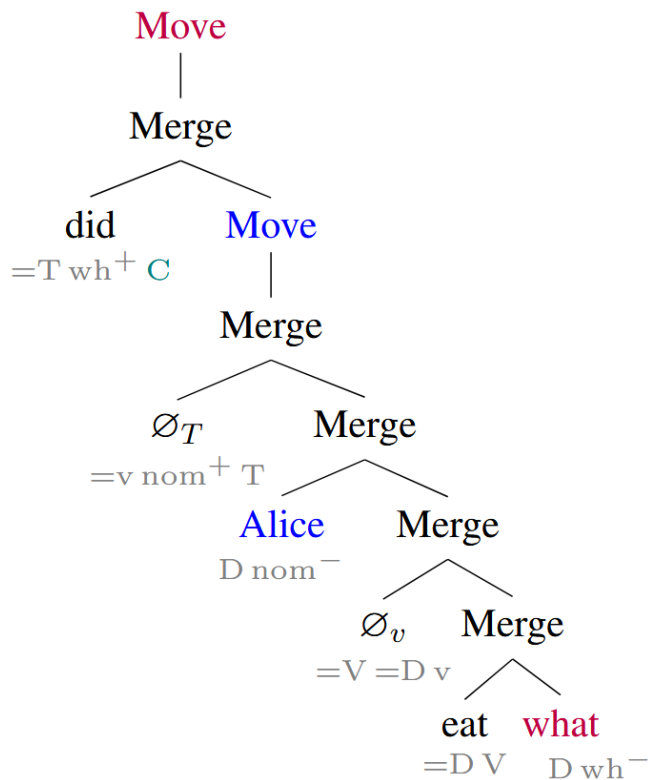
Example



Example



Example



Discussion and Conclusion

Discussion and Conclusion

- We have constructed a **tensor semantics for MGs**, showing how their combinatory operations can be used to **incorporate compositionality into distributional semantics**

Discussion and Conclusion

- We have constructed a **tensor semantics for MGs**, showing how their combinatory operations can be used to **incorporate compositionality into distributional semantics**
- Our approach is **directly compatible with existing tensor-based approaches** to MG syntax (beim Graben & Gerth, 2012)

Discussion and Conclusion

- We have constructed a **tensor semantics for MGs**, showing how their combinatory operations can be used to **incorporate compositionality into distributional semantics**
- Our approach is **directly compatible with existing tensor-based approaches** to MG syntax (beim Graben & Gerth, 2012)
- Ideas for future work

Discussion and Conclusion

- We have constructed a **tensor semantics for MGs**, showing how their combinatory operations can be used to **incorporate compositionality into distributional semantics**
- Our approach is **directly compatible with existing tensor-based approaches** to MG syntax (beim Graben & Gerth, 2012)
- Ideas for future work
 - See how the vector embeddings can be **learned from data**

Discussion and Conclusion

- We have constructed a **tensor semantics for MGs**, showing how their combinatory operations can be used to **incorporate compositionality into distributional semantics**
- Our approach is **directly compatible with existing tensor-based approaches** to MG syntax (beim Graben & Gerth, 2012)
- Ideas for future work
 - See how the vector embeddings can be **learned from data**
 - Versions of **Move** which are **not semantically inert** and incorporate semantic ideas such as variable binding, etc.

Discussion and Conclusion

- We have constructed a **tensor semantics for MGs**, showing how their combinatory operations can be used to **incorporate compositionality into distributional semantics**
- Our approach is **directly compatible with existing tensor-based approaches** to MG syntax (beim Graben & Gerth, 2012)
- Ideas for future work
 - See how the vector embeddings can be **learned from data**
 - Versions of **Move** which are **not semantically inert** and incorporate semantic ideas such as variable binding, etc.

Overall, this is a step towards unifying connectionist and symbolic approaches to linguistic representations.

Thank you very much!

Paper:



<https://openpublishing.library.umass.edu/scil/article/id/4032/>

Slides:



https://github.com/JPJ280/jpj280.github.io/tree/master/publications/SCIL_2026_MG_Tensors/slides

References

- Peter beim Graben and Sabrina Gerth. 2012. Geometric representations for minimalist grammars. *Journal of logic, language, and information*, 21(4):393–432.
- Bob Coecke, Mehrnoosh Sadrzadeh, and Stephen Clark. 2010. Mathematical foundations for a compositional distributional model of meaning. *Linguistic Analysis*, 36:345–384.
- Aniello De Santo. 2020. Structure and memory: A computational model of storage, gradience, and priming. Ph.D. thesis, State University of New York at Stony Brook.
- Aniello De Santo. 2025. Capturing online SRC/ORC effort with memory measures from a minimalist parser. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*, pages 24–35, Albuquerque, New Mexico, USA. Association for Computational Linguistics.
- Gregory M. Kobele, Sabrina Gerth, and John Hale. 2012. Memory resource allocation in top-down minimalist parsing. In *Formal Grammar - 17th and 18th International Conferences, FG 2012, Opole, Poland, August 2012, Revised Selected Papers, FG 2013, Düsseldorf, Germany, August 2013. Proceedings, Lecture Notes in Computer Science*, pages 32–51. Springer.
- Jean Maillard, Stephen Clark, and Edward Grefenstette. 2014. A type-driven tensor-based semantics for CCG. In *Proceedings of the EACL 2014 Workshop on Type Theory and Natural Language Semantics (TTNLS)*, pages 46–54, Gothenburg, Sweden. Association for Computational Linguistics.
- Mark Steedman. 2001. *The syntactic process*. MIT press.
- Mark Steedman and Jason Baldridge. 2011. Combinatory categorial grammar. In Robert D. Borsley and Kersti Börjars, editors, *Non-Transformational Syntax*, chapter 5, pages 181–224. John Wiley & Sons, Ltd.

Construction of Φ

Selectee Features in our MG	Corresponding Vector Space
D	D
V	V
v	U
T	T
C	C

- $\mathbf{S} := \mathbf{D} \oplus \mathbf{V} \oplus \mathbf{U} \oplus \mathbf{T} \oplus \mathbf{C}$
- $\mathbf{L} := \bigoplus_{k \geq 1} \mathbf{S}^{\otimes k}$
 - $\mathbf{S}^{\otimes k}$ means $\underbrace{\mathbf{S} \otimes \mathbf{S} \otimes \dots \otimes \mathbf{S}}_{k \text{ times}}$
- Lex is the lexicon over the MG
- T_A is the set of trees over the MG
- $\varphi: Lex \rightarrow \mathbf{L}$ maps each lexical item to the space corresponding to its feature string
- $\Phi: T_A \rightarrow \mathbf{L}$ extends φ by
 - $\Phi(t) = \varphi(t)$, if $t \in Lex$
 - $\Phi(\text{merge}(t_1, t_2)) = \Phi(t_1) \cdot \Phi(t_2)$
 - $\Phi(\text{move}(t)) = \Phi(t)$

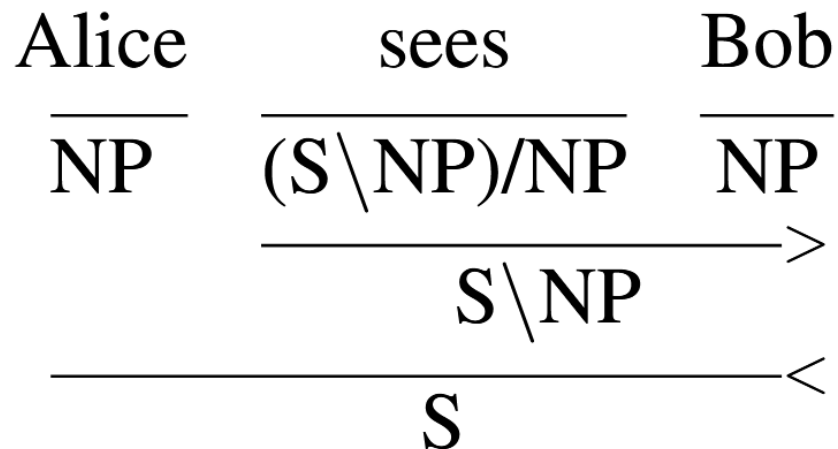
Tensor Contraction (General)

- Let $A \in U_1 \otimes \cdots \otimes U_l \otimes V_1 \otimes \cdots \otimes V_m$ be an order $l + m$ tensor
- Let $B \in V_1 \otimes \cdots \otimes V_m \otimes W_1 \otimes \cdots \otimes W_n$ an order $m + n$ tensor
- Denote by $a_{i_1 \dots i_l j_1 \dots j_m}$ the element at index $i_1, \dots, i_l, j_1, \dots, j_m$ of A (thought of as an $l + m$ dimensional grid of numbers); likewise for $b_{j_1 \dots j_m k_1 \dots k_n}$
- The *tensor contraction* of A and B (along the indices $j_1 \dots j_m$) is an order $l + n$ tensor $AB \in U_1 \otimes \cdots \otimes U_l \otimes W_1 \otimes \cdots \otimes W_n$, with the element at $i_1, \dots, i_l, k_1, \dots, k_n$ given by

$$\sum_{j_1=1}^{\dim(V_1)} \cdots \sum_{j_m=1}^{\dim(V_m)} a_{i_1 \dots i_l j_1 \dots j_m} b_{j_1 \dots j_m k_1 \dots k_n}$$

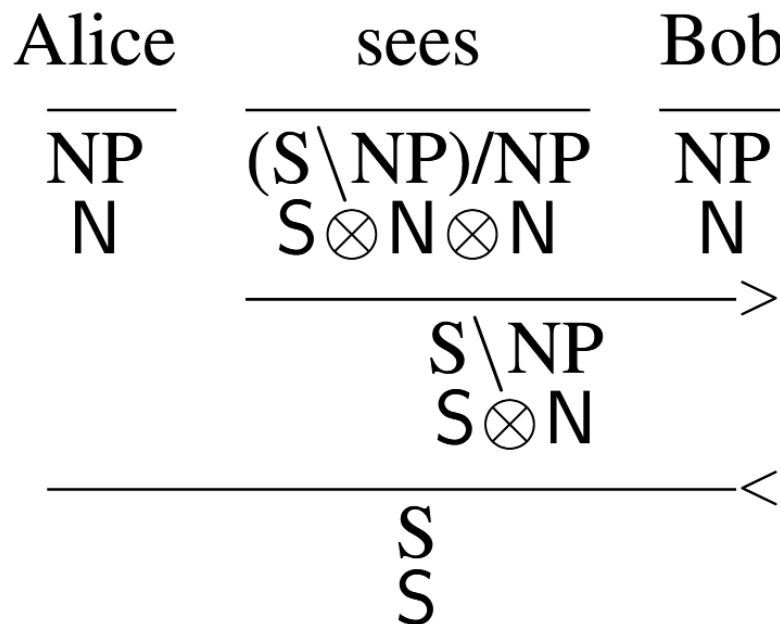
Combinatory Categorical Grammar (CCG)

- Mildly Context Sensitive formalism
- Thought to be capable of expressing all human syntax
- Steedman (2001), Steedman and Baldridge (2011)



CCG and Tensor Semantics

- Approach due to Maillard et al. (2014), Coecke et al. (2010), a.o.



beim Graben and Gerth (2012)

- $\mathcal{F}$ is a vector space
- T_A is the set of trees over an MG
- **merge**: $\mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}$ is a partial binary operation
- **move**: $\mathcal{F} \rightarrow \mathcal{F}$ is a partial unary operation
- $\Psi: T_A \rightarrow \mathcal{F}$ is a function such that
 - $\Psi(\text{merge}(t_1, t_2)) = \text{merge}(\Psi(t_1), \Psi(t_2))$
 - $\Psi(\text{move}(t)) = \text{move}(\Psi(t))$
- Gives a geometric vector space representation for the syntax of MGs

Our Semantics with bG&G's Syntax

- $\Omega: T_A \rightarrow (\mathcal{F} \times \mathbf{L}), t \mapsto (\Psi(t), \Phi(t))$
- $\mathcal{M}erge: (\mathcal{F} \times \mathbf{L}) \times (\mathcal{F} \times \mathbf{L}) \rightarrow (\mathcal{F} \times \mathbf{L}), ((\mathbf{a}, A), (\mathbf{b}, B)) \mapsto (\mathbf{merge}(\mathbf{a}, \mathbf{b}), A \cdot B)$
 - $\Omega(\mathbf{merge}(t_1, t_2)) = \mathcal{M}erge(\Omega(t_1), \Omega(t_2))$
- $\mathcal{M}ove: (\mathcal{F} \times \mathbf{L}) \rightarrow (\mathcal{F} \times \mathbf{L}), (\mathbf{a}, A) \mapsto (\mathbf{move}(\mathbf{a}), A)$
 - $\Omega(\mathbf{move}(t)) = \mathcal{M}ove(\Omega(t))$